

Link Diagnosis Enabled by Longitudinal Noise Monitoring

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Abstract We show that a longitudinal noise monitoring scheme utilizing coherent receiver DSP comprehensively monitors link health. By experimentally demonstrating dynamic SNR monitoring during channel-drop transients and the localization of abnormal noise increases, we establish an effective, spatially resolved diagnostic tool. ©2026 The Author(s)

Introduction

The emergence of coherent optical receivers as sophisticated multi-purpose sensors has revolutionized modern network telemetry. Traditionally tasked with data recovery, these receivers now enable longitudinal power monitoring (LPM) or power profile estimation (PPE), providing unprecedented visibility into the physical layer [1,2]. By leveraging the digital signal processing (DSP) capabilities inherent in coherent detection, operators can extract critical channel parameters—such as power-vs-distance profiles [3,4], polarization-dependent loss (PDL) [5], differential group delay (DGD) [6], and optical filter [7] characteristics distributedly—without the need for expensive, standalone optical time-domain reflectometers (OTDRs), or other instruments along the link.

Building upon this foundation, we recently introduced an advanced LPM scheme for monitoring the signal-to-noise ratio (SNR) distributed along the fiber link [8]. Unlike conventional LPM algorithms that provide only the power related characterization, our technique provides insight into the SNR along the signal path, offering a high-fidelity "health map" of the optical path.

As networks transition toward more dynamic, disaggregated, and open architectures, the ability to localize impairments becomes a prerequisite for autonomous management. However, beyond standard performance monitoring, there is a growing urgency to address complex signal integrity issues and physical-layer security threats that evade traditional detection. These include channel add/drop induced transients, multi-path interference (MPI) caused by connector reflections, inter modal interference (IMI) in hollow core fiber, in-band crosstalk from low port isolation of the wavelength selective switch (WSS), and malicious noise injection attacks designed to degrade quality of service (QoS) or mask data interception.

In this paper, we extend our longitudinal noise monitoring technology to the specialized domain of link anomaly detection. We demonstrate that

by analyzing the longitudinal noise evolution, we can distinguish between benign signal degradation and specific "invisible" link issues. This work presents two types of applications of the longitudinal noise monitoring scheme: one is the temporal evolution of the survival channels SNR during channel add/drop; another one is the detection of abnormal noise increase due to faults, such as MPI, in-band crosstalk, and noise injection attack.

Methodology: Longitudinal Noise Monitoring

In dispersion-unmanaged transmission links, the non-commutativity between chromatic dispersion and the Kerr effect generates unique local nonlinear distortions (NLDs) that are captured by the coherent receiver. Conventional LPM detects these by digitally regenerating the NLDs using the transmitted (Tx) waveform, typically reconstructed from post-FEC or hard-decision symbols. By correlating these noise-free regenerated NLDs with received NLDs, conventional LPM enables the longitudinal estimation of the signal power alone, insensitive to accumulated noise power.

We recently proposed an advanced method to monitor distributed longitudinal noise along the fibre link [8]. In this approach, we utilize the total received waveform to generate local NLDs. Since the total waveform comprises both signal and noise components, the resulting NLDs incorporate the NLDs of both. Correlating these noise-inclusive NLDs with the received signal enables the longitudinal monitoring of the total power (signal plus noise) evolution. Furthermore, by combining conventional LPM with our proposed noise monitoring method, the longitudinal SNR can be estimated.

Monitoring Transient Events: Temporal Evolution of Channel SNR

In our previous work, we demonstrated the capability of fast and accurate LPM to capture time-transient events, such as channel loading changes during a fast add/drop scenario [9]. That demonstration achieved a monitoring rate of up

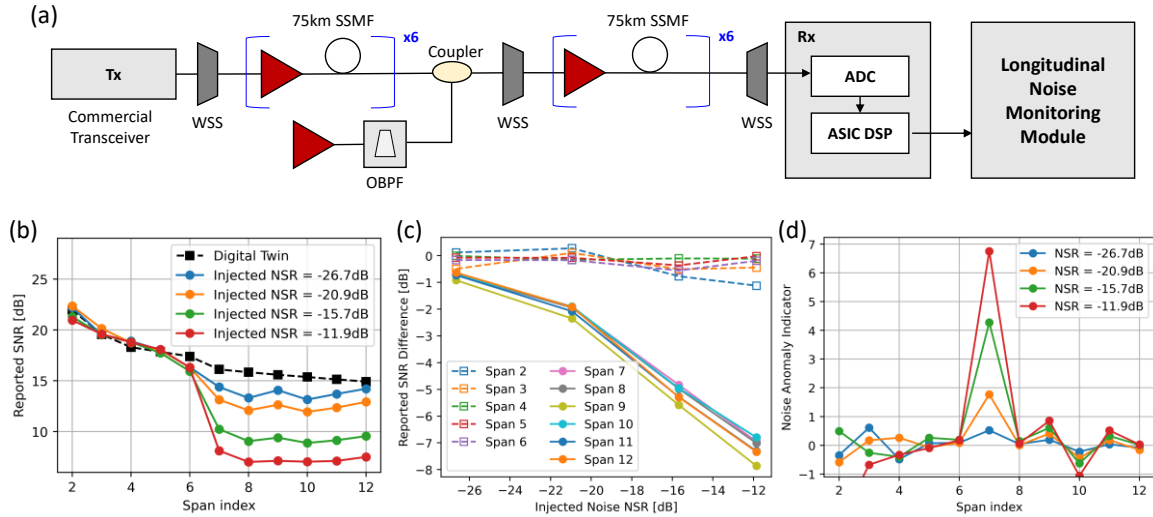


Fig. 2: (a) Experimental setup for detection and localization of abnormal noise increases. (b) Reported SNR profiles obtained via longitudinal noise monitoring method at varying noise injection levels. (c) Reported SNR variation as a function of injected noise NSR. (d) Calculated noise anomaly indicator identifying the spatial location of the abnormal noise increases.

existence of these impairments is usually unknown, so they cannot be modelled through a standard digital twin. Consequently, while monitoring these hidden noise increases is critical for network health, they are very challenging to detect. Our proposed longitudinal method detects all the noise accumulated in the link; therefore, it can provide an effective way to monitor these noise increases.

As an example of the abnormal noise degradation, we emulate a noise injection attack using the experimental setup illustrated in Fig. 2(a). A commercial Tx was used to generate a 138-Gbaud DP-QPSK signal centred at 194.1 THz. The signal was transmitted across 12 spans of SSMF, with each span measuring 75 km and the launch power maintained at 6 dBm per span. A 99:1 optical coupler was placed at the end of the 6th span to inject ASE noise into the link. This ASE noise was shaped using an optical bandpass filter to match the spectrum of the transmitted signal, while a variable optical attenuator controlled the injected power level. After transmission, the signal was captured by a commercial Rx, and the final waveform was recovered using the receiver's built-in DSP. This output waveform was then fed into the longitudinal noise monitoring module. To reduce stochastic error, a total of 50M symbols were used to generate the final trace.

Fig. 2(b) illustrates the reported SNR under varying levels of injected ASE noise power. The SNR at each span input, estimated using a digital twin, is represented by the black curve. A digital twin estimates SNR using static system parameters (e.g., amplifier noise figures, insertion losses, span launch powers), it does not account for

anomalous injected noise, rendering the noise injection invisible. In contrast, the reported longitudinal SNR profiles reveal a distinct and abrupt drop precisely at the noise injection point, while remaining unchanged in the preceding spans across all noise levels. The variation in the reported SNR as a function of injected noise power is detailed in Fig. 2(c). The first half of the link (prior to injection) is plotted using dashed curves, while the second half (post-injection) is shown with solid curves. These SNR variations are normalized against a baseline reference profile taken without noise injection. As expected, the SNR of the spans before the noise injection remains constant regardless of the varying ASE noise, whereas the proportional SNR degradation is observed in the spans after the injection location. Moreover, comparing the reported SNR to a reference profile (either pre-measured or digital-twin-simulated) yields the noise anomaly indicator shown in Fig. 2(d). A distinct peak emerges exactly at the noise injection location, confirming abnormal noise detection and localization.

Conclusions

Monitoring noise within a transmission link is critical for network management. Until now, there has been a lack of effective monitoring methods, particularly those capable of fault localization. In this paper, we experimentally demonstrate two practical applications of our proposed longitudinal noise monitoring method. The first is monitoring the temporal evolution of link noise during a fast channel drop event and the second is the detection and localization of abnormal noise increase. The proposed distributed approach has broad application potential beyond these two scenarios, offering a vital, location-resolved tool for mapping overall link health.

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